Lesson 1

INTRODUCTION TO MICROSOFT EXCEL

Excel in Business

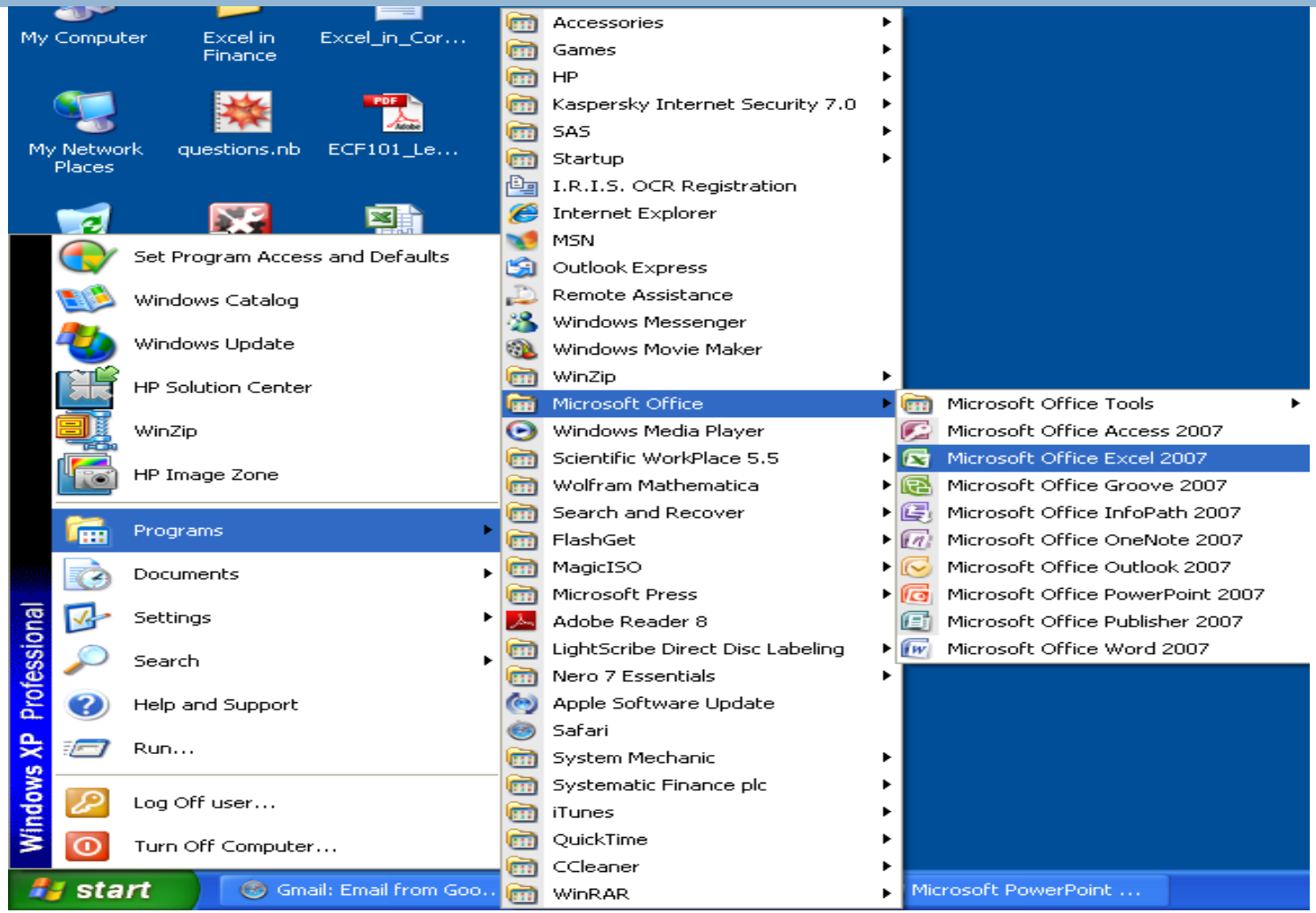
What do you need?

- Microsoft Excel installed on your Windows system (as part of Microsoft Office)

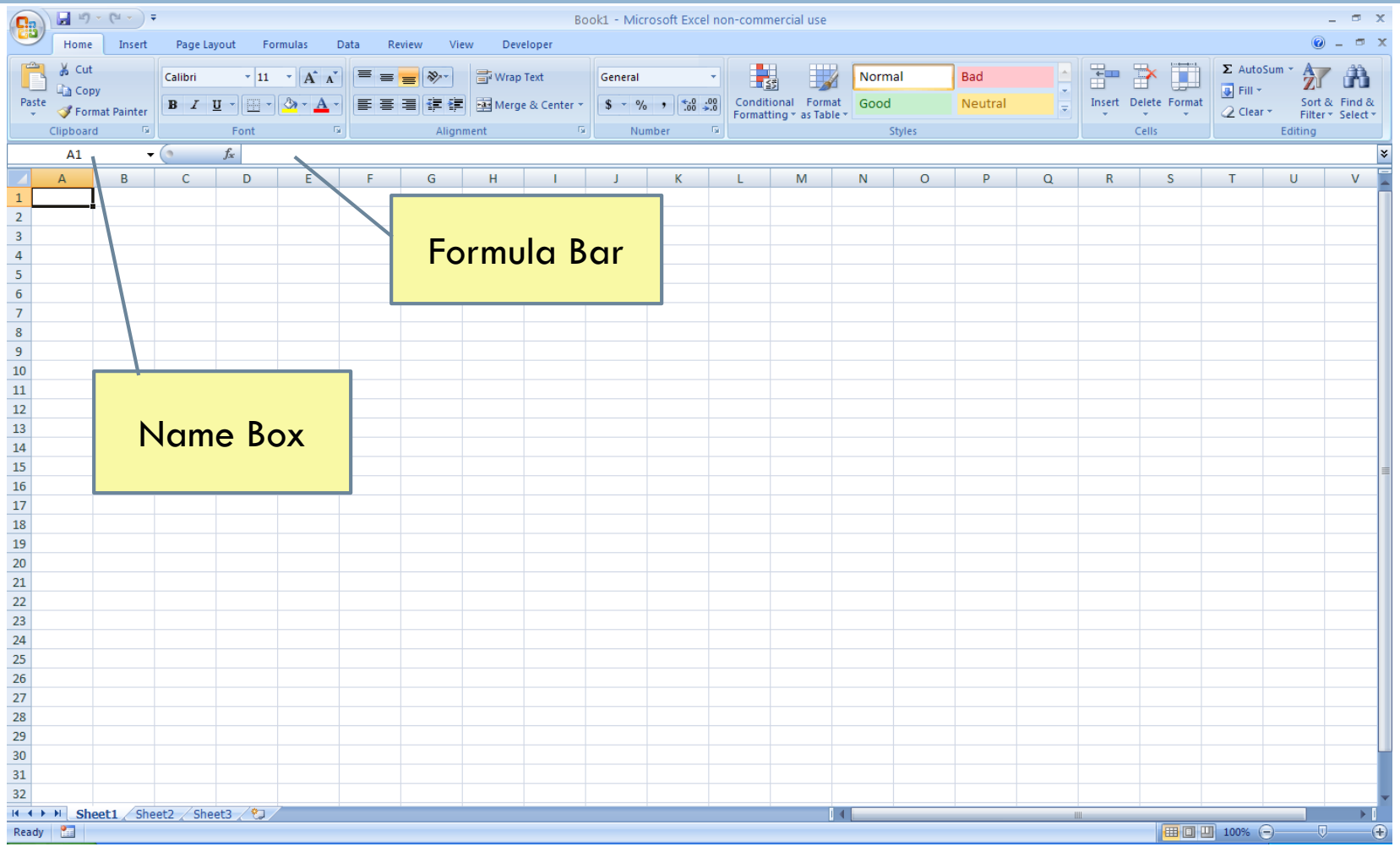
What is Microsoft Excel?

- A powerful spreadsheet program that
  - Performs numerical computations
  - Can organize/summarize huge data sets
  - Carries out advanced statistical and financial analysis

Where is it located?



A First Look at Excel



Points to Note

- Rows are numbered as 1,2,3,…
- Columns are marked as A,B,C,…
- Name Box always displays the current selected cell (i.e., where the cursor is positioned)

Excel as a Desk Calculator

You can perform simple operations like addition, multiplication and division.

Simply select a cell and in the formula bar enter the expression. Begin the expression with an equal to (=) sign.

To compute 5+7 you have to enter **=5+7** and then press the **Enter** key.

Various Arithmetic Operations

Operation	Symbol
Addition	+
Subtraction	-
Multiplication	*
Division	/
Raise to power	^

Exercises

Calculate the following:

4+5x(4-2)	
4+5x4-2	
$\frac{1}{2^3} \div \frac{1}{3^2}$	
2^{40}	

Exercises

Calculate the following:

4+5x(4-2)	14
4+5x4-2	22
$\frac{1}{2^3} \div \frac{1}{3^2}$	1.125
2^{40}	1.09951E+12

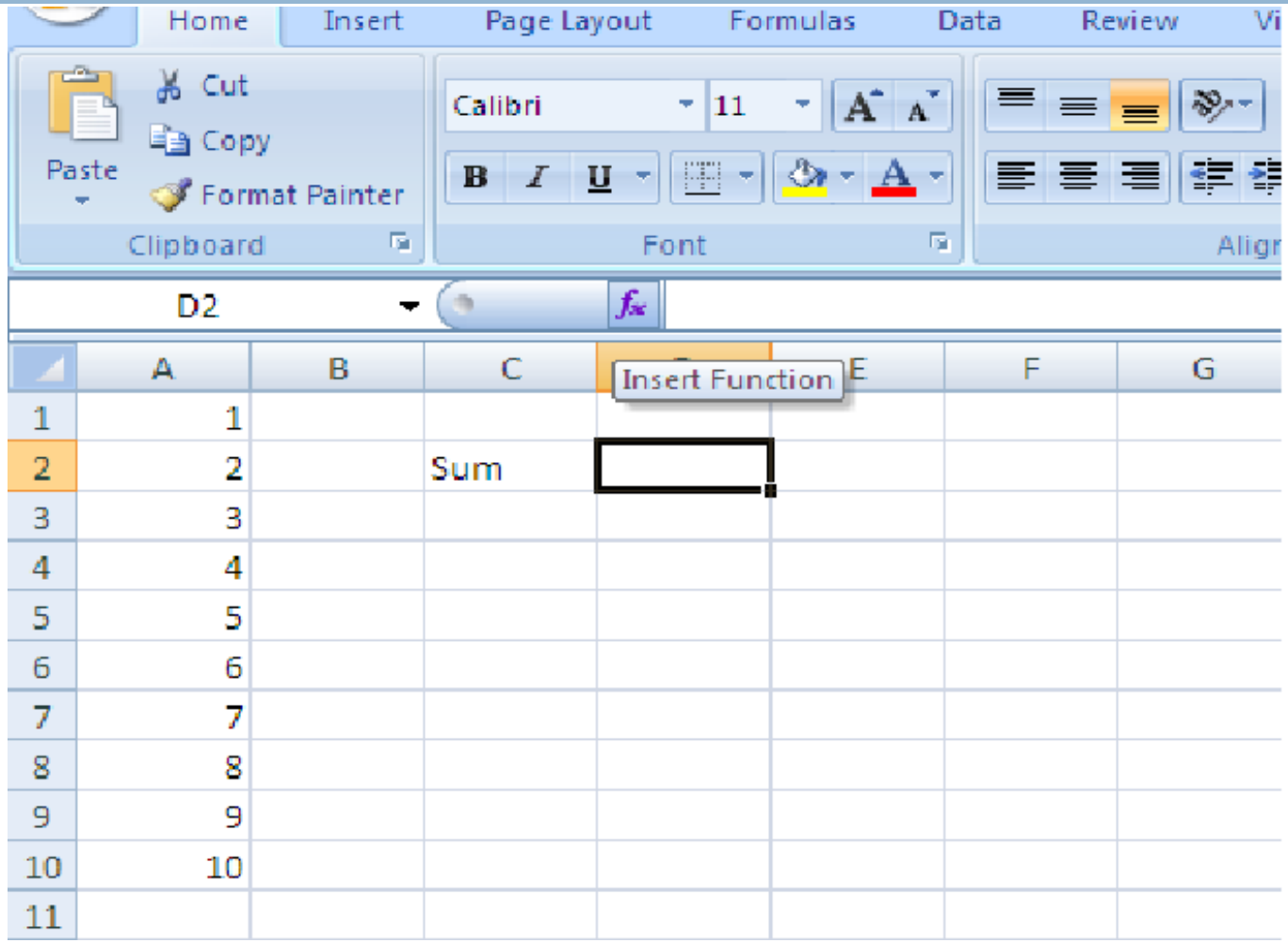
SUM Function

Computes the sum of a list of numbers

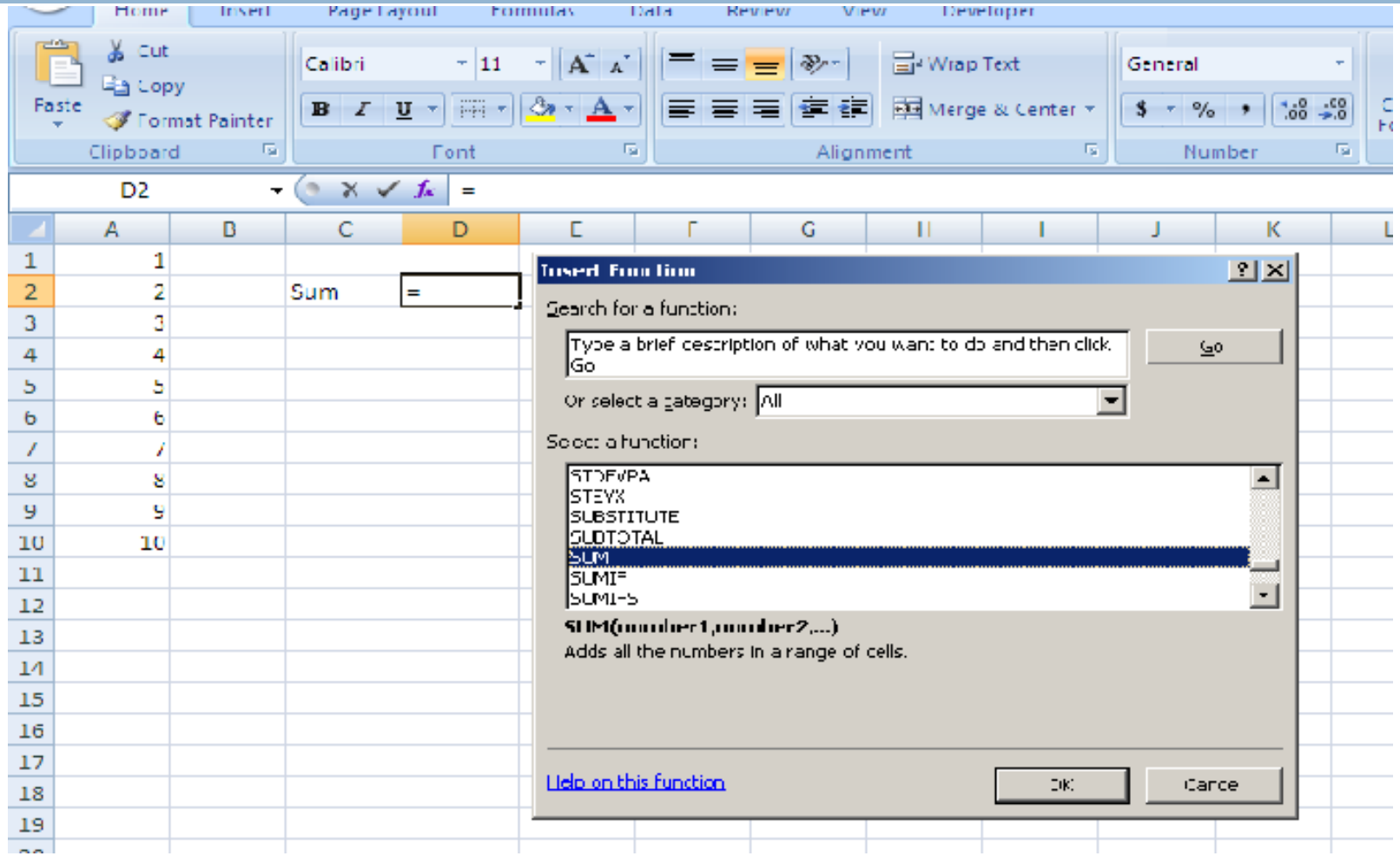
Syntax Example: **=SUM(A1:A10)**

where A1:A10 is the range that stores the numbers. This range could be of any size.

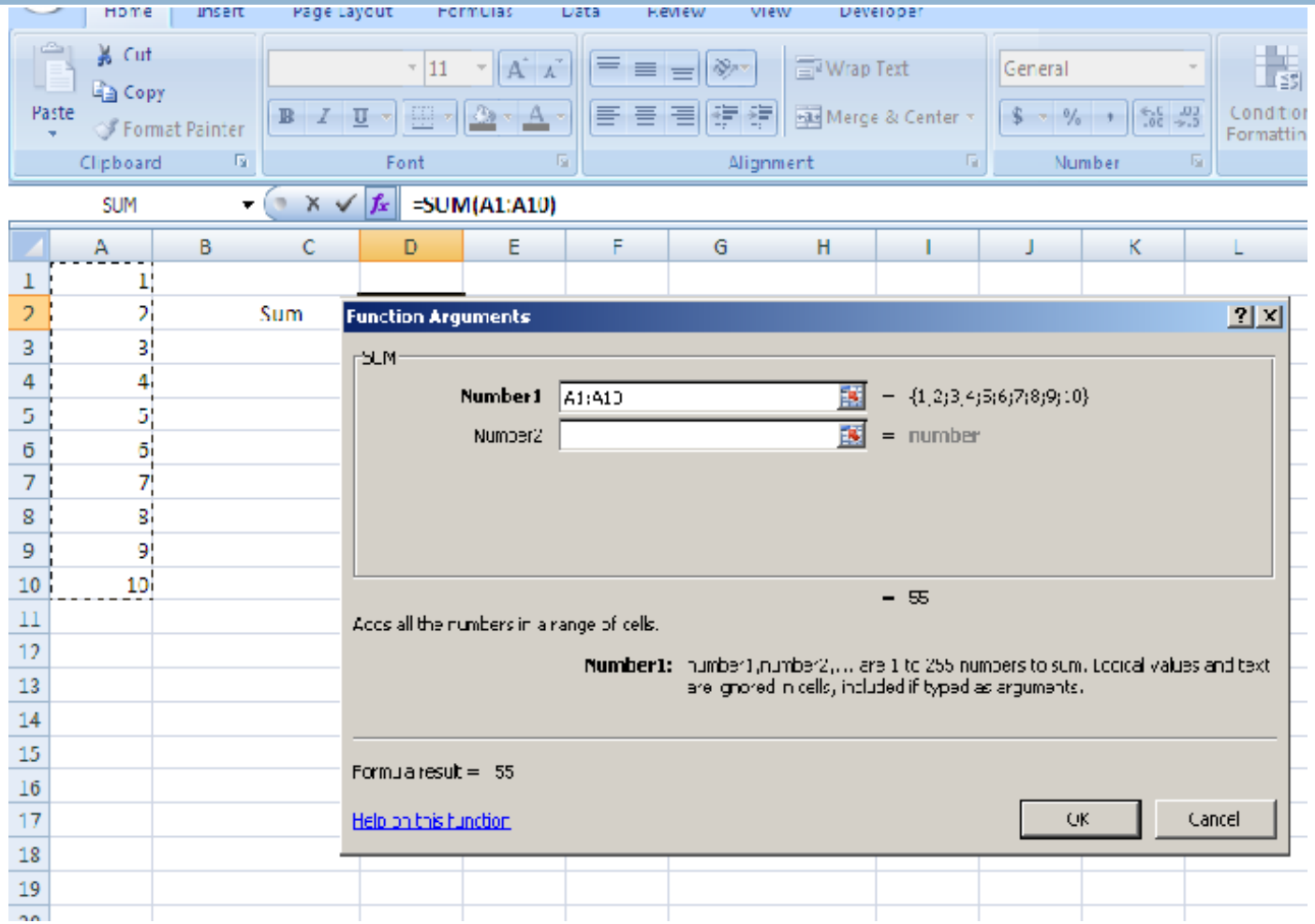
Built-in Functions



Built-in Functions



Built-in Functions



Calculating Data Properties

AVERAGE Function

- Computes the arithmetic mean

- Syntax Example: **=AVERAGE(B1:B50)**

  where B1:B50 is the range that stores the numbers. This range could be of any size.

AVERAGE – some remarks

- The arguments must either be numbers, arrays or references that contain numbers.

- If an array or reference argument contains text, or empty cells, those values are ignored.

AVERAGE – an exercise

- Apply AVERAGE to each of the following data columns

A	B	C	D
5	5	5	5
13		0	X
15	15	15	15

AVERAGEA

- In addition to numbers, text and logical values such as TRUE and FALSE are also included in the calculation.
- All text is treated as Zero.
- Except – TRUE is treated as One.

AVERAGEA – an exercise

- Apply AVERAGEA to each of the following data columns

A	B	C	D	E
5	5	5	5	TRUE
13		0	X	TRUE
15	15	15	15	FALSE

A Few Useful Functions

- SQRT
- COS, SIN, TAN
- EXP
- LOG10, LN
- ROUND
- COUNT
- MAX, MIN
- MEDIAN
- MODE
- PERCENTILE
- QUARTILE
- TRIMMEAN

PERCENTILE

- **Percentiles** are the numbers which divide the ordered data set in 100 equal sized data subsets.
- For instance, $P_{0.95}$, the **95 % percentile** is a number such that at most 95% of the data points are less than it and at most 5% of the data points are greater than it.

PERCENTILE function

- **PERCENTILE(range of data, K)**
- **K** is a number between 0 and 1
- The PERCENTILE function gives a cutoff with 100K% of the data below it.

PERCENTRANK function

- Conversely, we take a particular data point and ask for the corresponding percentile. The syntax for this is:

- PERCENTRANK(range of data, data point)

QUARTILES

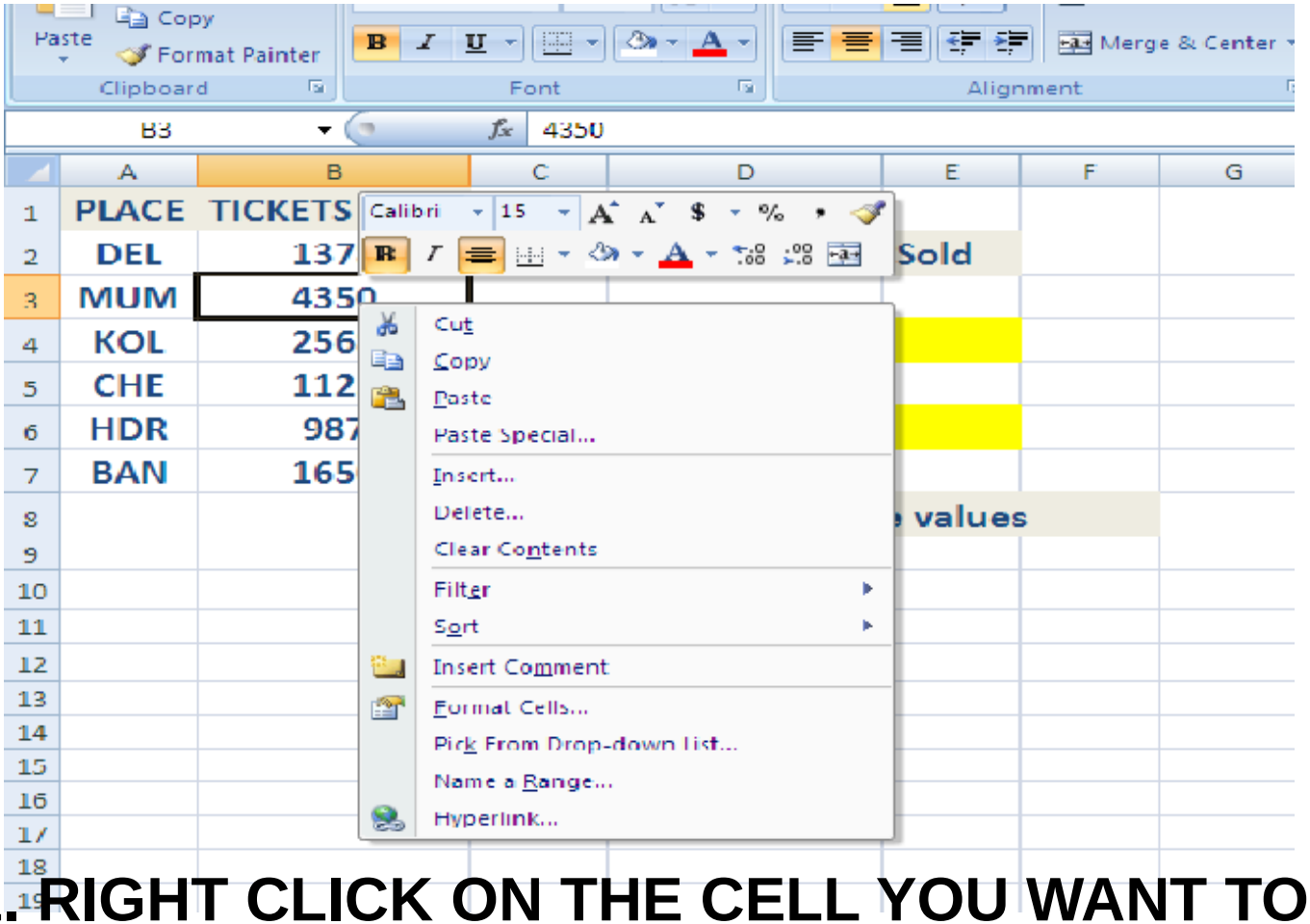
- The 25%, 50% and 75% percentiles cut the data into four equal parts – these percentiles are known as the first, second and third quartiles.

- QUARTILE(range of data, q)

  where q = 1, 2 or 3.

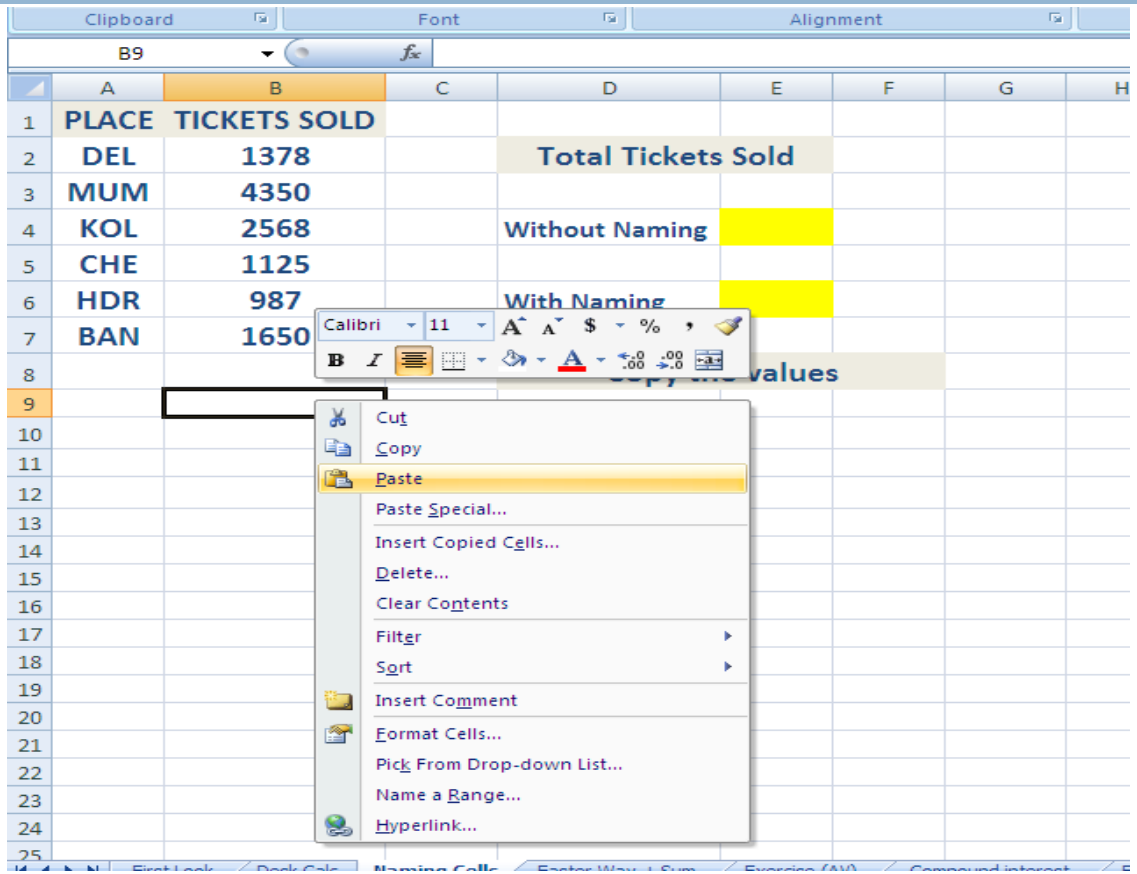
Working with Cells

Copying from One Cell to Another



1. RIGHT CLICK ON THE CELL YOU WANT TO COPY

Copying from One Cell to Another



2. RIGHT CLICK ON THE CELL WHERE YOU WANT TO PASTE

Copying a Function

- Enter "=SUM(A2:A4)" in cell A1
- Copy the content of cell A1 and paste in cell B5
- Select B5 and look at its content.

What do you see?

B5 shows "SUM(B6:B8)" !!!

Static Cell References

- To keep a cell reference from changing use $ signs.
- Enter "=SUM($A$2:$A$4)" in cell A1
- Copy the content of cell A1 and paste in cell B5
- Select B5 and look at its content.

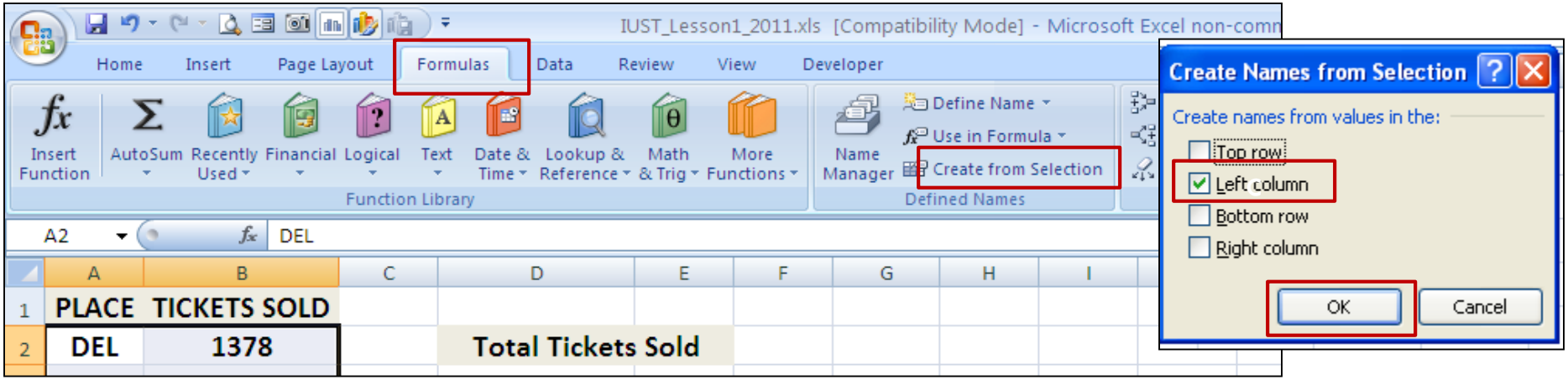
Now B5 shows "SUM($A$2:$A$4)"

Naming Cells – Method 1

- Cells can be given names
- Open the “Naming Cells” worksheet.
- Select B2. In the Name Box type “del”. Now B2 has the name “del”.
- Similarly, name B3 to B7 “mum”, “kol”, etc.
- These cells can now be called by these names.
- This is also a way to have static cell references.

Naming Cells – Method 2

- Open the "Naming Cells" worksheet.
- Select cells A2:B7.
- In FORMULAS tab, DEFINED NAMES group, select CREATE FROM SELECTION.
- Select LEFT COLUMN and press OK.



Some Useful Tips

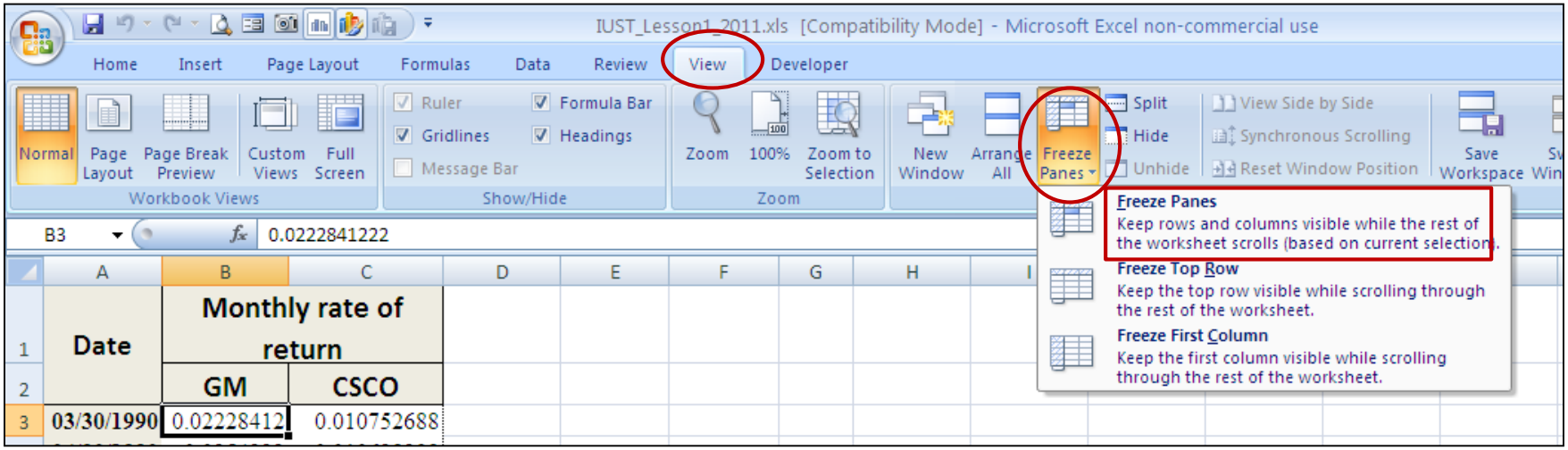
Freeze Panes in Excel

- You can lock rows or columns in one area by freezing or splitting panes.

- When you freeze panes, you select specific rows or columns that remain visible when scrolling in the worksheet.

Freeze Panes in Excel: An example

- Go to worksheet Histogram and select cell B3.
- In VIEW tab, WINDOW group, select FREEZE PANES.
- Select FREEZE PANES.



View multiple worksheets at the same time

To View 2 worksheets in the same workbook side by side.

- In VIEW tab, WINDOW group, select NEW WINDOW.
- In VIEW tab, WINDOW group, select VIEW SIDE BY SIDE.
- To scroll both worksheets at the same time, click SYNCHRONOUS SCROLLING in the Window group on the View tab.

View multiple worksheets at the same time

To View 2 worksheets of different workbooks side by side.

- Open both the workbooks that contain the worksheets to compare.
- In VIEW tab, WINDOW group, select VIEW SIDE BY SIDE.
- To scroll both worksheets at the same time, click SYNCHRONOUS SCROLLING in the Window group on the View tab.

View multiple worksheets at the same time

To View multiple worksheets at the same time.

- Open all the workbooks that contain the worksheets to compare.
- In VIEW tab, WINDOW group, select ARRANGE ALL.
- Under ARRANGE, click the option that you want.